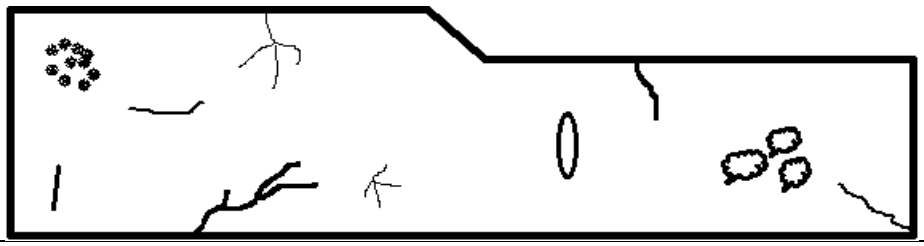
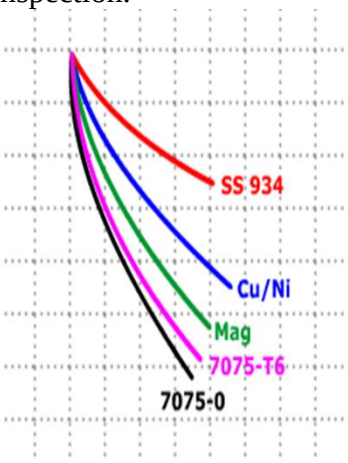
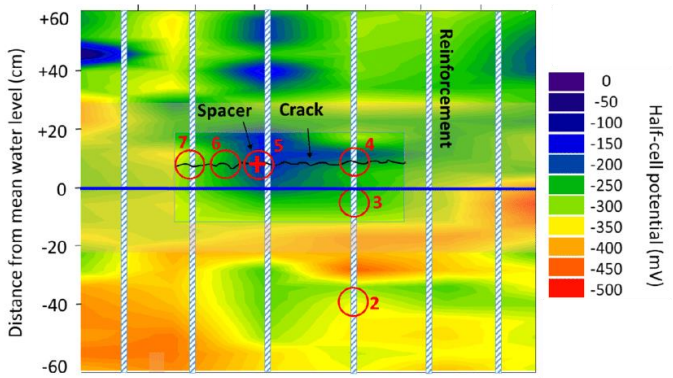
MANIPAL INSTITUTE OF TECHNOLOGY
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DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION - REGULAR		**November, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**PE - VII**
Course Name	**Non-destructive Testing (NDT) of Concrete Structures**	**Course Code**	**CIE 4080**
Semester	**VII**	**Max Marks**	**50**
Date	**20/11/2024**	**Duration**	**3 hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	A concrete column is inspected by subjecting the core to compression and measuring its strength. At the same time, the column was also assessed using the rebound hammer method. Compare the aforementioned methods.	5	1	1,2,6, 12	4
1B	Explain any two disadvantages of each NDT method used to detect subsurface defects in concrete.	3	1	1,2,6, 12	3
1C	Explain the characteristics and functions of the penetrant used in liquid penetrant inspection (LPI).	2	2	5,6,9, 12	3
2A	Illustrate, with a neat sketch, the procedure for determining surface defects in steel bridge girder using post-emulsifiable lipophilic LPI.	5	2	5,6,9, 12	3
2B	Illustrate the working principle of the ultrasound probe with a neat sketch.	3	2	5,6,9, 12	3
2C	Illustrate the process of applying dry particles used in magnetic particle inspection (MPI).	2	3	5,8,12	3
3A	With a neat sketch, explain the normal probe reflection technique of ultrasound inspection for detecting surface and sub-surface defects present in metals.	5	3	5,8,12	3
3B	Explain the procedure for determining the depth of carbonation in concrete.	3	3	5,8,12	2
3C	Explain the role of the differential coil used in the eddy current inspection technique.	2	3	5,8,12	2

4A	The defects shown in the diagram below were inspected using ultrasound. Mention the type of probe and probe technique to be used for this purpose. Sketch a typical waveform for each type of defect, and explain the procedure for inspection of defects. 	**5**	5	**1,2,3, 4,12**	**4**
4B	Analyse the impedance plane diagram, shown below, obtained during materials sorting using eddy current inspection. 	**3**	5	**1,2,3,4 ,12**	**4**
4C	Explain how reflection strength, positive reflection polarity, and negative reflection polarity of ground penetrating radar (GPR) affect the interpretation of results?	**2**	5	**1,2,3,4 ,12**	**3**
5A	A reinforced concrete column was inspected for any corrosion in reinforcement using half-cell potential, and the result obtained is presented in the figure below. Analyse the given figure for level of corrosion. 	**5**	5	**1,2,3,4 ,12**	**4**

5B	Analyse the defects in concrete structure as shown in the below figures.  **(a)**  **(b)**  **(c)**	**3**	5	**1,2,3,4 ,12**	**4**
5C	Explain the effect of the type of aggregate on the pulse velocity of ultrasound and the compressive strength of concrete.	**2**	5	**1,2,3,4 ,12**	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***8***
2	***2***
3	***18***
4	***0***
5	***22***

BL	Total Marks
2	**5**
3	**24**
4	**21**
5	**0**